SOLAR/DUNE Significance Analyses

Mathematical Derivations: Day-Night, HEP, and Sensitivity

SOLAR/DUNE Analysis

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1 Overview and Scientific Goals

This document presents the mathematical derivations underlying the three significance analyses in the SOLAR/DUNE software framework, ordered from simplest to most complex:

1. **Day-Night Asymmetry** (`12DayNight.py`): searches for a time-modulated excess of solar neutrinos during nighttime relative to daytime, driven by the MSW matter effect inside the Earth. Significance is computed from both a Gaussian approximation on the rate-difference spectrum and a two-sample Poisson log-likelihood ratio (Asimov).

2. **HEP Discovery** (`13HEP.py`): searches for the hep solar neutrino flux ($^3\mathrm{He} + p \to {}^4\mathrm{He} + e^+ + \nu_e$) as an absolute excess above background. Three significance estimators are computed: Gaussian, Asimov, and a profile-likelihood (PL) significance with a single global background nuisance, analytically profiled.

3. **Sensitivity** (`14Sensitivity.py`): maps the 2D sensitivity contour in the oscillation-parameter plane (Δm^2, $\sin^2\theta_{12}$). A Baker-Cousins Poisson deviance with two normalization nuisances is minimized numerically over 2D templates in energy and azimuth.

The three analyses share common ingredients (histogram smoothing, thresholds, adaptive rebinning for HEP) but differ in the statistical complexity of their model. Table 1 (Section 6) summarizes the key differences.

Common notation. Throughout, T is the exposure in kt·yr, M_{det} the active detector mass in kt, and the exposure factor is $\mathcal{E} = T\,M_{\mathrm{det}}$. Index i runs over energy bins; index j over azimuth bins (Sensitivity only). Component label c identifies one physical process (signal, neutron, gamma, radiological, ^{8}B).

2 Common Ingredients

2.1 Rate Histograms and Exposure Scaling

All three analyses work with per-unit-exposure, per-unit-mass rate histograms r_i^c (units: $(\mathrm{yr}\cdot\mathrm{kt}\cdot\mathrm{MeV})^{-1}$, integrated over the bin width ΔE). Expected event counts at exposure $\mathcal{E}$ are:

$$\mu_i^c = \mathcal{E}\,r_i^c = T\,M_{\mathrm{det}}\,r_i^c. \tag{1}$$

All analyses apply a reconstructed-energy threshold E_{th} (configured in `analysis/config.json`); only bins with $E_i \geq E_{\mathrm{th}}$ enter the significance computation.

2.2 Histogram Smoothing

Each rate histogram r_i^c is convolved with a one-dimensional Gaussian kernel before entering the significance computation:

$$\tilde{r}_i^c = \sum_j G_\sigma(i-j)\,r_j^c, \qquad G_\sigma(k) = \frac{1}{\sqrt{2\pi}\,\sigma}\exp\left(-\frac{k^2}{2\sigma^2}\right), \tag{2}$$

implemented via `scipy.ndimage.gaussian_filter1d` with `mode='nearest'`. Whether smoothing is applied to a given component, and the width σ, are configured in `analysis/smoothing.jso` under `SMOOTHING.ANALYSES.{ANALYSIS}.STAGES`. Separate stages are provided for the fiducial scan and the significance computation.

Non-negativity clipping. Gaussian convolution at distribution tails can produce small negative values. Negative rates are unphysical and cause profile-likelihood divergences at high exposure (Section 4.5.5). All smoothed rates are therefore clipped to zero before any significance computation:

$$\tilde{r}_i^c \leftarrow \max(0,\, \tilde{r}_i^c)\,. \tag{3}$$

2.3 Per-Component Background Error Model

For each background component c the absolute uncertainty on the raw rate in bin i combines MC-statistical and systematic contributions in quadrature:

$$\sigma_i^c = r_i^c \sqrt{\left(\frac{\delta_i^c}{r_i^c}\right)^2 + (\sigma_{\rm rel}^c)^2}, \tag{4}$$

where δ_i^c/r_i^c is the relative histogram statistical error and $\sigma_{\rm rel}^c$ is the relative systematic uncertainty (`--background_uncertainty`, typically 2%). The total combined background uncertainty in bin i is:

$$\sigma_i^{\rm bkg} = \sqrt{\sum_c (\sigma_i^c)^2}, \tag{5}$$

also convolved with the component-specific smoothing kernel. For signal components (hep, ^{8}B) the corresponding quantity uses the signal systematic $\sigma_{\rm rel}^{\rm sig} = 30\%$ (HEP) or 4% (Sensitivity).

3 Day-Night Asymmetry Analysis

3.1 Physical Observable: MSW-Enhanced Day-Night Effect

Solar ν_e produced in the solar core oscillate in vacuum on their way to Earth. During nighttime passages the neutrinos traverse the Earth's interior and experience additional matter-induced (MSW) oscillations that partially restore the ν_e flavour component. This creates a night-time excess characterised by the asymmetry:

$$\Delta_{\rm DN} \equiv \frac{N_{\rm night} - N_{\rm day}}{\frac{1}{2}(N_{\rm night} + N_{\rm day})}. \tag{6}$$

The size of $\Delta_{\rm DN}$ depends on the Earth electron-density profile through the MSW potential $V = \sqrt{2}\, G_F\, n_e(r)$ and on the oscillation parameters (Δm^2, $\sin^2\theta_{12}$).

3.2 Two-Period Signal and Background Model

Let $f \in (0,1)$ be the fraction of total exposure attributed to daytime and $g = 1 - f$ the nighttime fraction ($f = 0.5$ by default; the SURF latitude of 44.35°N gives $f \approx 0.493$ averaged over a full year). The oscillation-weighted solar rates split into day and night components:

$$r_i^{\rm day} = \text{solar rate in daytime sky exposure}, \tag{7}$$

$$r_i^{\rm night} = \text{solar rate in nighttime (Earth-crossing) exposure}. \tag{8}$$

Isotropic backgrounds (cosmogenic, geological, detector-intrinsic) are assumed time-uniform and therefore split proportionally between periods.

Event counts observed in day and night periods are:

$$n_i^{\text{night}} = \mathcal{E}\big(g\, r_i^{\text{bkg}} + r_i^{\text{night}}\big), \tag{9}$$

$$n_i^{\text{day}} = \mathcal{E}\big(f\, r_i^{\text{bkg}} + r_i^{\text{day}}\big). \tag{10}$$

The **asymmetry signal** at scale factor ξ (Section 3.7) is:

$$s_i = \mathcal{E} \cdot \xi \cdot \big(r_i^{\text{night}} - r_i^{\text{day}}\big). \tag{11}$$

3.3 Effective Two-Sample Background

For a two-sample rate-difference test with unequal period lengths f and g, the optimal significance statistic requires an *inverse-fraction-weighted* effective background:

$$B_i^{\text{eff}} = \frac{n_i^{\text{night}}}{g^2} + \frac{n_i^{\text{day}}}{f^2}. \tag{12}$$

In the equal-period limit $f = g = \frac{1}{2}$ and with $s_i \ll B_i^{\text{eff}}$, this reduces to $B_i^{\text{eff}} \approx 4\mathcal{E}r_i^{\text{bkg}}$ and the Gaussian significance becomes the familiar $s_i/\sqrt{b_i}$ form. Bins where $B_i^{\text{eff}} = 0$ are zeroed.

3.4 Gaussian Significance

Two Gaussian estimators are computed per bin.

Simple Gaussian (no additional background uncertainty):

$$Z_i^{\text{G}} = \frac{s_i}{\sqrt{B_i^{\text{eff}}}}, \tag{13}$$

Error Gaussian (with combined background uncertainty σ_i^{eff}, see Section 3.5):

$$Z_i^{\text{G,err}} = \frac{s_i}{\sqrt{B_i^{\text{eff}} + \big(\sigma_i^{\text{eff}}\big)^2}}, \tag{14}$$

The global significance is the quadrature sum over all bins above threshold:

$$Z = \sqrt{\sum_{i \geq i_{\text{th}}} (Z_i)^2}. \tag{15}$$

Both raw and Gaussian-smoothed rate versions are used, producing four output curves per cut per asymmetry-scale value (`RawGaussian`, `RawErrorGaussian`, `Gaussian`, `ErrorGaussian`) plus upper/lower band variants (Section 3.7).

3.5 Background Uncertainty for the Rate-Difference Test

Three independent noise sources are combined in quadrature per period and per bin, then merged into an effective uncertainty:

1. **Poisson statistical**: $\sqrt{N_{\text{bkg},\,p}}$, where $N_{\text{bkg},\,p} = \mathcal{E}\, f_p\, r_i^{\text{bkg}}$ is the background count in period $p \in \{\text{day}, \text{night}\}$.
2. **Normalization systematic**: $\sigma_{\text{rel}}^{\text{bkg}}\, N_{\text{bkg},\,p}$ (`--background_uncertainty`).
3. **Day-fraction systematic**: $\delta_f\, \mathcal{E}\, r_i^{\text{bkg}}$ (`--day_fraction_band`, $\delta_f = 0.01$ by default), reflecting uncertainty in the solar zenith angle cut and the run schedule.

Per-period uncertainties:

$$\sigma_{p,i} = \sqrt{N_{\text{bkg},\,p,i} + \left(\sigma_{\text{rel}}^{\text{bkg}}\, N_{\text{bkg},\,p,i}\right)^2 + \left(\delta_f\, \mathcal{E}\, r_i^{\text{bkg}}\right)^2}. \tag{16}$$

Combining across periods via inverse-fraction weighting (mirroring Eq. (12)):

$$\sigma_i^{\text{eff}} = \sqrt{\left(\frac{\sigma_{\text{night},i}}{g}\right)^2 + \left(\frac{\sigma_{\text{day},i}}{f}\right)^2}, \tag{17}$$

set to zero wherever $B_i^{\text{eff}} = 0$.

3.6 Two-Sample Poisson Asimov Significance

The Day-Night analysis also computes a two-sample Poisson log-likelihood ratio (Asimov) as an alternative to the Gaussian estimators. Under the signal hypothesis (asymmetry scale ξ), the Asimov (expected) observed counts are n_i^{night} and n_i^{day} from Eqs. (9)–(10). The null hypothesis H_0 assumes no asymmetry: both periods share the same per-bin total rate, allocated proportionally to period length:

$$h_{0,i}^{\text{night}} = g\,(n_i^{\text{night}} + n_i^{\text{day}}), \qquad h_{0,i}^{\text{day}} = f\,(n_i^{\text{night}} + n_i^{\text{day}}). \tag{18}$$

The two-sample log-likelihood ratio is:

$$q_i = 2\left[n_i^{\text{night}} \ln \frac{n_i^{\text{night}}}{h_{0,i}^{\text{night}}} + n_i^{\text{day}} \ln \frac{n_i^{\text{day}}}{h_{0,i}^{\text{day}}}\right], \tag{19}$$

defined as zero whenever any count or null hypothesis count is non-positive. The global Asimov significance is:

$$Z_A = \sqrt{\max\left(0, \sum_{i \geq i_{\text{th}}} q_i\right)}. \tag{20}$$

This is the natural two-sample extension of the HEP Asimov formula (Section 4.4).

3.7 Asymmetry Uncertainty Band

Two independent sources of uncertainty on the predicted asymmetry amplitude are combined in quadrature to define a total band ε_{tot}:

1. **Earth density band** $\varepsilon_{\oplus}$ (`--earth_density_band`, default 0.13): fractional uncertainty from PREM-based oscillation probability calculations.
2. **Oscillation parameter band** ε_{osc} (`--oscillation_band`, default 0.05): residual uncertainty from PDG θ_{12} and Δm_{21}^2 ranges.

$$\varepsilon_{\mathrm{tot}} = \sqrt{\varepsilon_{\oplus}^2 + \varepsilon_{\mathrm{osc}}^2} \approx 0.138 \text{ (defaults)}. \tag{21}$$

Three scale factors bracket the full predicted range:

$$\xi \in \{1+\varepsilon_{\mathrm{tot}},\ 1,\ 1-\varepsilon_{\mathrm{tot}}\}. \tag{22}$$

The scale acts on the asymmetry signal only (Eq. (11)), leaving the total event rate and the background unchanged. This produces three significance curves (upper, nominal, lower) for each estimator.

3.8 Results

Figures 1–4 show the Day-Night analysis outputs for both HD configurations using the SolarEnergy estimator at the nominal selection.

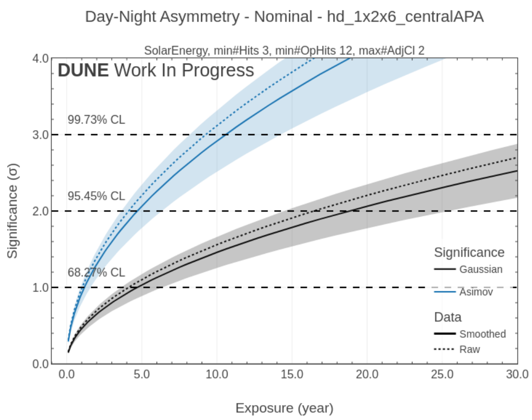

Figure 1: centralAPA: exposure needed for 3σ and 5σ discovery as a function of day-night asymmetry magnitude.

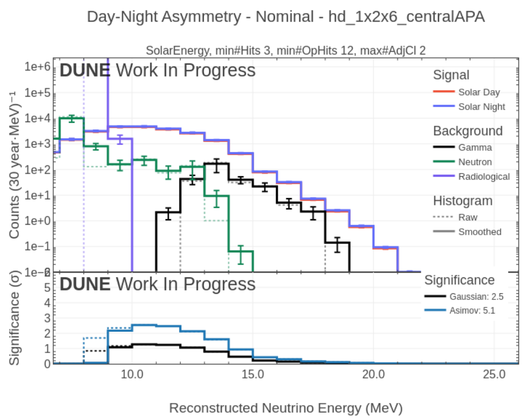

Figure 2: centralAPA: significance vs. exposure at 30 kt·yr for nominal, upper, and lower asymmetry bands.

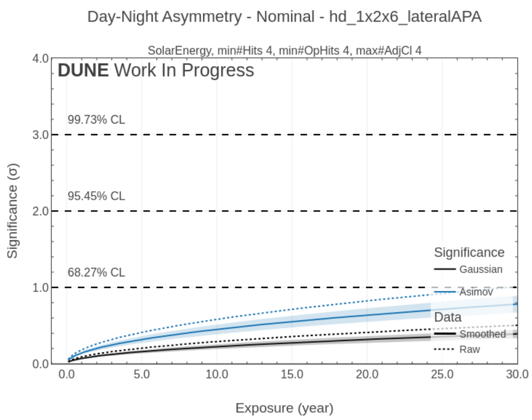

Figure 3: lateralAPA: exposure threshold for 3σ and 5σ Day-Night discovery.

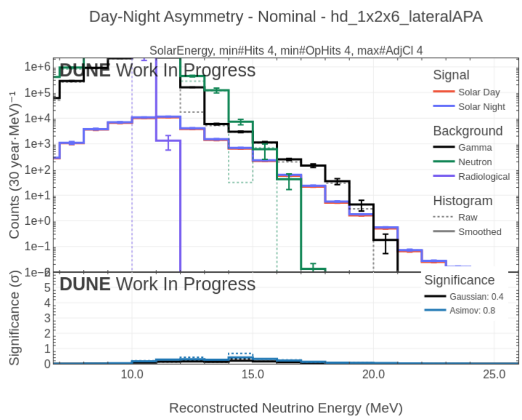

Figure 4: lateralAPA: significance vs. exposure at 30 kt·yr.

4 HEP Discovery Analysis

4.1 Signal and Background Model

The HEP analysis searches for an absolute rate excess of hep solar neutrinos above a known background in a one-dimensional energy spectrum above E_{th}. Signal (hep) and total background event counts in bin i are:

$$s_i = \mathcal{E}\, r_i^{\mathrm{hep}}, \tag{23}$$

$$b_i = \mathcal{E} \sum_{c \neq \mathrm{hep}} r_i^c = \mathcal{E}\big(r_i^\gamma + r_i^n + r_i^{\mathrm{rad}} + r_i^{^8\mathrm{B}}\big). \tag{24}$$

The Asimov dataset sets observed counts to $n_i = s_i + b_i$.

4.2 Background Error Aggregation

Each background component c contributes an absolute uncertainty σ_i^c (Eq. (4)). The combined error and its smoothed counterpart enter the significance computation as:

$$\sigma_i^{\mathrm{bkg}} = \sqrt{\sum_{c \neq \mathrm{hep}} \left(\sigma_i^c\right)^2}. \tag{25}$$

4.3 Adaptive Tail Rebinning

4.3.1 Detection Threshold

A bin (or merged group) is declared *detectable* if the expected (conservatively shifted) signal exceeds:

$$T_{\mathrm{det}} = \max(T_{\mathrm{min}},\ -\ln(1-p_{\mathrm{min}}))\,, \tag{26}$$

with defaults $T_{\mathrm{min}} = 1.0$ and $p_{\mathrm{min}} = 1-e^{-1}$, giving $T_{\mathrm{det}} = 1.0$. The Poisson interpretation: $P(\geq 1; \lambda) \geq p \Leftrightarrow \lambda \geq -\ln(1-p)$.

The conservative (downward-fluctuated) detectable signal is:

$$s_i^{\mathrm{det}} = s_i\,(1 - d \cdot \sigma_{\mathrm{rel}}^{\mathrm{sig}}), \quad d \in \{2.9,\ 3.0,\ 3.1\}, \tag{27}$$

where the three values of d correspond to the upper (+), nominal, and lower (−) error bands respectively.

4.3.2 Greedy Algorithm and Monotonicity Enforcement

Bins are merged from the high-energy tail inward until each group exceeds T_{det}. Groups below T_{det} are zeroed out. To guarantee a non-decreasing significance vs. exposure curve, if the current step's optimal binning yields lower Asimov significance than the previous step the previous binning is retained:

$$\mathcal{B}(t) = \begin{cases} \mathcal{B}^*(t) & Z_A(\mathcal{B}^*(t), t) \geq Z_A(\mathcal{B}(t-1), t-1), \\ \mathcal{B}(t-1) & \text{otherwise.} \end{cases} \tag{28}$$

4.4 Gaussian and Asimov Significance

With s_i and b_i from Eqs. (23)–(24) and combined background uncertainty $\sigma_i^{\rm bkg}$, the per-bin Gaussian estimator is:

$$Z_i^{\rm G} = \frac{s_i}{\sqrt{b_i + \left(\sigma_i^{\rm bkg}\right)^2}}, \tag{29}$$

and the global significance is Eq. (15).

The Asimov significance [1], with and without background uncertainty, is:

$$Z_A = \sqrt{2\left[(s_i + b_i)\ln\left(1 + \frac{s_i}{b_i}\right) - s_i\right]}, \tag{30}$$

$$Z_A^\sigma = \sqrt{2\left[(s_i + b_i)\ln\frac{(s_i + b_i)(b_i + (\sigma_i^{\rm bkg})^2)}{b_i^2 + (s_i + b_i)(\sigma_i^{\rm bkg})^2} - \frac{b_i^2}{(\sigma_i^{\rm bkg})^2}\ln\left(1 + \frac{(\sigma_i^{\rm bkg})^2 s_i}{b_i(b_i + (\sigma_i^{\rm bkg})^2)}\right)\right]}. \tag{31}$$

These are computed both with and without adaptive tail rebinning. The profile-likelihood (Section 4.5) is the primary metric.

4.5 Profile-Likelihood Significance

4.5.1 Test Statistic

The profile-likelihood significance follows Cowan et al. [1]. The test statistic for the null hypothesis $\mu = 0$ (no HEP signal) is:

$$q_0 = -2\ln\frac{\mathcal{L}(0,\,\hat{\beta})}{\mathcal{L}(\hat{s} + b,\,1)}, \tag{32}$$

evaluated at the Asimov point $n_i = s_i + b_i$, with $\hat{\beta}$ the background scale factor profiled under the null. The median discovery significance is $Z = \sqrt{q_0}$.

4.5.2 Global Background Normalization Nuisance

A single global background scale factor β — fully correlated across all bins — is constrained by a Gaussian penalty:

$$\mathcal{L}(0, \beta) = \prod_{i=1}^{K} \frac{(\beta b_i)^{n_i} e^{-\beta b_i}}{n_i!} \cdot \exp\left(-\frac{(\beta - 1)^2}{2\sigma_{\rm rel}^2}\right), \tag{33}$$

where $\sigma_{\rm rel}$ is the relative background uncertainty (`--background_uncertainty`, typically 2%). A global (fully correlated) β avoids the K-parameter absorbing plateau that per-bin nuisances produce; the plateau ends at $T \sim 1/(B_{\rm tot}\sigma_{\rm rel}^2)$ (typically sub-year), giving a physically correct PL curve.

4.5.3 Profiling $\hat{\beta}$

The stationarity condition $\partial \ln \mathcal{L}/\partial\beta\big|_{\hat{\beta}} = 0$ at the Asimov point $n_i = s_i + b_i$ gives:

$$N_{\rm tot} = \hat{\beta} B_{\rm tot} + \frac{\hat{\beta} - 1}{\sigma_{\rm rel}^2}, \quad N_{\rm tot} = \sum_i n_i,\ B_{\rm tot} = \sum_i b_i. \tag{34}$$

Because β is *global*, the per-bin denominators βb_i factorize and the stationarity condition is a scalar equation. Rearranging:

$$\hat{\beta}^2 + \left(B_{\text{tot}}\sigma_{\text{rel}}^2 - 1\right)\hat{\beta} - N_{\text{tot}}\sigma_{\text{rel}}^2 = 0, \tag{35}$$

with analytic positive root:

$$\hat{\beta} = \frac{-(B_{\text{tot}}\sigma_{\text{rel}}^2 - 1) + \sqrt{(B_{\text{tot}}\sigma_{\text{rel}}^2 - 1)^2 + 4N_{\text{tot}}\sigma_{\text{rel}}^2}}{2}. \tag{36}$$

When $B_{\text{tot}} = 0$ or $\sigma_{\text{rel}} = 0$, $\hat{\beta} = 1$ (no constraint active).

4.5.4 Log-Likelihood Ratio

Substituting $n_i = s_i + b_i$ into Eq. (32):

$$q_0 = 2\sum_{i=1}^{K}\left[n_i \ln\frac{n_i}{\hat{\beta}b_i} - (n_i - \hat{\beta}b_i)\right] + \left(\frac{\hat{\beta}-1}{\sigma_{\text{rel}}}\right)^2. \tag{37}$$

Each per-bin term is the Baker-Cousins Poisson deviance [2] between the Asimov observation n_i and the null expectation $\hat{\beta}b_i$. This expression — a sum of Poisson deviances with a global nuisance penalty — is the structural prototype for the Sensitivity objective (Section 5.4), extended to 2D with two nuisances.

4.5.5 Numerical Stability

When $S^2/B \ll 1$, naive subtraction of two $\mathcal{O}(N \log N)$ log-likelihoods loses precision. The stable per-bin form is:

$$\Delta\ell_i = n_i \ln\frac{n_i}{\mu_i^{\text{null}}} - \left(n_i - \mu_i^{\text{null}}\right), \qquad \mu_i^{\text{null}} = \hat{\beta}b_i, \tag{38}$$

with a floor $\epsilon_{\min} = 10^{-12}$ replacing zero denominators.

Negative-rate blowup. If a smoothed rate is negative, $\hat{\beta}b_i < 0$, the floor gives $\Delta\ell_i \approx n_i \ln(n_i/\epsilon_{\min}) \propto T$, causing a spurious superlinear spike. Clipping (Eq. (3)) eliminates this path entirely.

4.6 Barlow-Beeston MC Mask

Bins with summed background MC count below $N_{\text{MC}}^{\min}$ (default 1.0) lack a reliable background model and are excluded by a static binary mask [3]:

$$\mathcal{M}_i = \mathbf{1}\left[N_i^{\text{MC}} \geq N_{\text{MC}}^{\min}\right], \qquad s_i \leftarrow \mathcal{M}_i s_i, \quad b_i \leftarrow \mathcal{M}_i b_i. \tag{39}$$

The mask is static (MC counts do not change with exposure), so it introduces no discrete transitions in the significance curve. The same mask is applied to both raw and smoothed histograms.

4.7 Profile-Likelihood Signal Bands and Post-Processing

Three signal normalizations are evaluated to produce $\pm 1\sigma_s$ bands, using the same scale factor $d \in \{+1, 0, -1\}$ applied to the signal:

$$s_i^{(d)} = s_i\,(1 + d \cdot \sigma_{\text{rel}}^{\text{sig}}). \tag{40}$$

Background is never shifted; $\hat{\beta}$ and its quadratic solution are unaffected.

The raw PL significance curve $Z(T)$ is post-processed by two steps when `pl_isotonic` is enabled (controlled by `analysis/config.json` under `WORKFLOW.HEP.pl_isotonic`):

1. **Gaussian kernel smoothing** with $\sigma_{\text{PL}} = 6$ exposure-grid index units (`scipy.ndimage.gaussian_fi` which suppresses numerical oscillations from the PL solver at low signal-to-background ratio.

2. **Isotonic regression (PAVA)** [4]:

$$\min_{\{z_k\}} \sum_k (z_k - \tilde{Z}(T_k))^2 \quad \text{s.t.} \quad z_k \leq z_{k+1}, \tag{41}$$

   via `sklearn.isotonic.IsotonicRegression`, which enforces a non-decreasing significance curve while minimising the ℓ^2 deviation from the computed values.

Pipeline: $Z \to \max(0, Z) \xrightarrow{\text{Gauss}} \tilde{Z} \xrightarrow{\text{PAVA}} Z'$. After isotonic regression, upper/lower bands are enforced to remain above/below the nominal curve respectively: $Z'^{(+1)} \leftarrow \max(Z'^{(+1)}, Z'^{(0)})$ and $Z'^{(-1)} \leftarrow \min(Z'^{(-1)}, Z'^{(0)})$. Pre-PAVA curves are saved separately for diagnostic purposes.

4.8 Spike Detection and Best-Cut Selection

A PL curve is declared *spiked* if any consecutive step in the pre-isotonic columns exceeds $\Delta_{\max}$ (`--max_pl_jump`, default $1\,\sigma$):

$$\text{spiked} \;\Leftrightarrow\; \max_k [Z_{\text{pre}}(T_{k+1}) - Z_{\text{pre}}(T_k)] > \Delta_{\max}. \tag{42}$$

The best cut maximises the reference significance over non-spiked cuts:

$$(N^*, N^*_{\text{op}}, N^*_{\text{adj}}) = \arg \max_{\text{not spiked}} Z_{\text{ref}}(T_{\max}). \tag{43}$$

4.9 Results

Figures 5–12 show HEP analysis outputs for both HD configurations (SolarEnergy, nominal). Each row presents: left, Gaussian/Asimov exposure comparison; right, profile-likelihood significance with adaptive rebinning.

5 Sensitivity Analysis

5.1 Overview

The Sensitivity analysis maps how well the detector can distinguish between the solar ($\Delta m^2_\odot$) and reactor ($\Delta m^2_{\text{react}}$) best-fit oscillation hypotheses. Rather than computing a

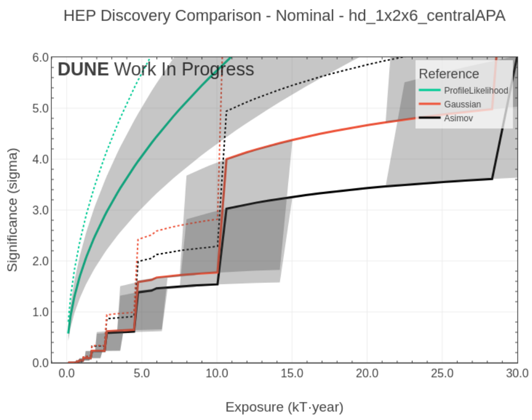

Figure 5: centralAPA: Gaussian and Asimov exposure needed to reach 3σ and 5σ hep discovery.

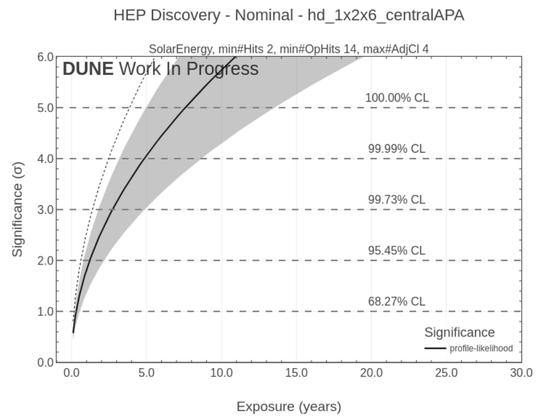

Figure 6: centralAPA: profile-likelihood significance vs. exposure.

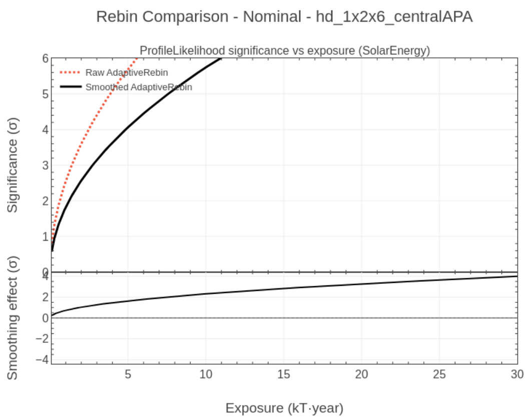

Figure 7: centralAPA: PL significance with and without adaptive rebinning.

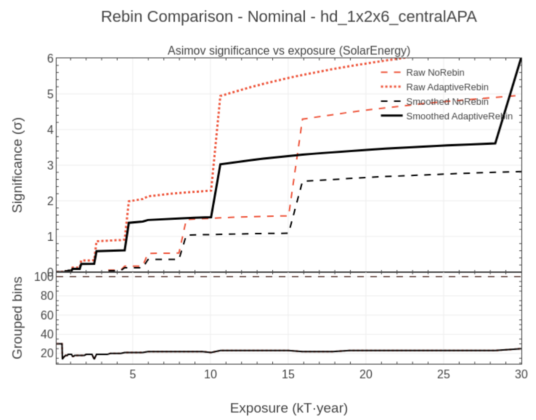

Figure 8: centralAPA: Asimov significance with and without adaptive rebinning.

single discovery significance it evaluates a χ^2 surface over the oscillation-parameter grid $(\Delta m^2, \sin^2\theta_{13}, \sin^2\theta_{12})$.

Key differences from HEP:

- Histograms are **two-dimensional** (energy $\times$ azimuth $\cos\eta$).
- **Two** normalization nuisances are floated: A_{pred} (signal) and A_{bkg} (background).
- The nuisance stationarity conditions have **no closed-form solution** — unlike HEP's quadratic for $\hat{\beta}$ — requiring full 2D numerical minimization.
- The goal is a χ^2 *map*, not a single significance.

5.2 Two-Dimensional Template Construction

Signal and background are represented as 2D arrays with axes energy $(j = 1, \ldots, J)$ and azimuth $(i = 1, \ldots, I)$.

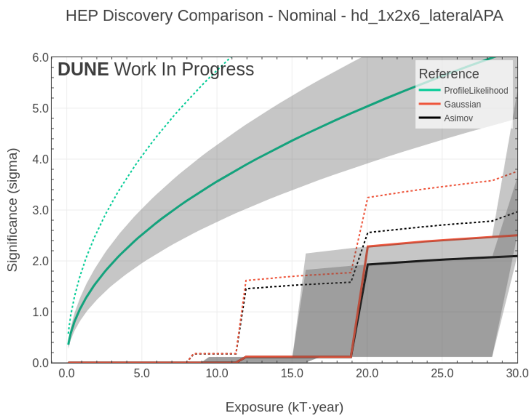

Figure 9: lateralAPA: Gaussian and Asimov exposure threshold for hep discovery.

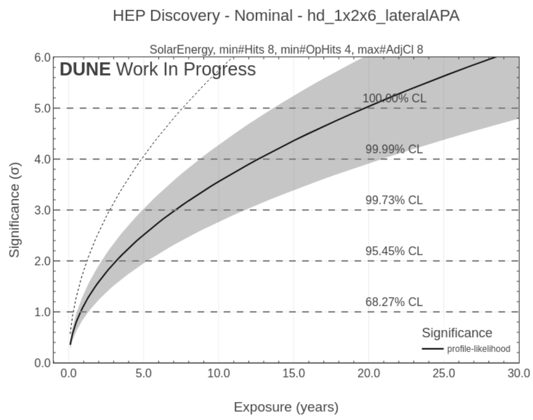

Figure 10: lateralAPA: profile-likelihood significance vs. exposure.

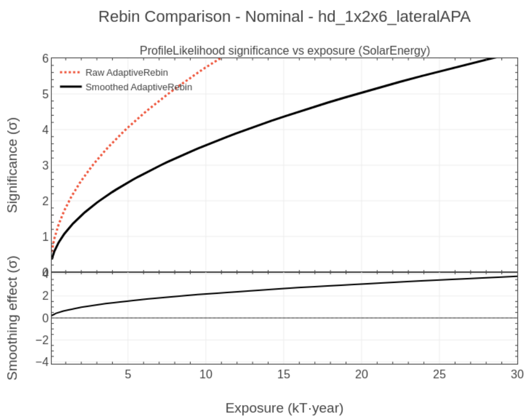

Figure 11: lateralAPA: PL significance with and without adaptive rebinning.

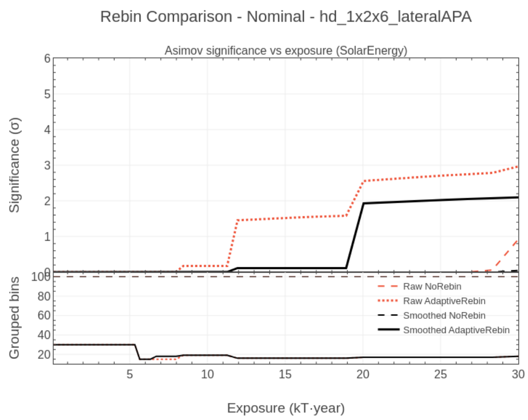

Figure 12: lateralAPA: Asimov significance with and without adaptive rebinning.

Signal template. For oscillation parameters $\vec{\theta} = (\Delta m^2, \sin^2\theta_{13}, \sin^2\theta_{12})$, the signal template at exposure $\mathcal{E}$ is:

$$T_{ij}^{\text{sig}}(\vec{\theta}) = \mathcal{E} \cdot M_{\text{det}} \cdot \left[P(\vec{\theta})\,H\right]_{ij}, \tag{44}$$

where $P(\vec{\theta})$ is the oscillation-probability matrix (azimuth $\times$ neutrino flavour) and H is the detector energy-response matrix (neutrino energy $\times$ reconstructed energy), implemented in `14SensitivitySignalTemplate.py`.

Background template. The background template T_{ij}^{bkg} is independent of oscillation parameters and is constructed from detector simulations in `14SensitivityBackgroundTemplate.py`.

5.3 Asimov Dataset Construction

For each scan point $\vec{\theta}_k$ the Asimov observed dataset is:

$$o_{ij}(\vec{\theta}_k) = T_{ij}^{\text{sig}}(\vec{\theta}_k) + T_{ij}^{\text{bkg}}. \tag{45}$$

Two *reference* templates fixed at the solar and reactor best-fit points:

$$p_{ij}^{\text{solar}} = T_{ij}^{\text{sig}}(\vec{\theta}_{\odot}), \qquad p_{ij}^{\text{react}} = T_{ij}^{\text{sig}}(\vec{\theta}_{\text{react}}). \tag{46}$$

5.4 Objective Function: Baker-Cousins Poisson Deviance

The fit minimizes a Baker-Cousins Poisson deviance [2] with two free normalization nuisances:

$$\chi^2(A_{\text{pred}}, A_{\text{bkg}}) = 2\sum_{i,j} \Delta\ell_{ij} + \left(\frac{A_{\text{pred}}}{\sigma_{\text{pred}}}\right)^2 + \left(\frac{A_{\text{bkg}}}{\sigma_{\text{bkg}}}\right)^2, \tag{47}$$

where the expected model is:

$$e_{ij} = (1 + A_{\text{bkg}})\, T_{ij}^{\text{bkg}} + (1 + A_{\text{pred}})\, p_{ij}, \tag{48}$$

and the per-bin Poisson deviance is:

$$\Delta\ell_{ij} = \begin{cases} e_{ij} - o_{ij} + o_{ij}\ln(o_{ij}/e_{ij}) & o_{ij} > 0,\ e_{ij} > 0, \\ e_{ij} & o_{ij} = 0,\ e_{ij} > 0, \\ 0 & e_{ij} = 0 \text{ (or masked)}. \end{cases} \tag{49}$$

Each term $\Delta\ell_{ij} \geq 0$ by the Gibbs inequality; the sum is zero if and only if $e_{ij} = o_{ij}$ for all bins.

Structural connection to HEP. Under H_0 (no signal, $p_{ij} = 0$), Eq. (47) reduces to the HEP log-likelihood ratio Eq. (37) with one nuisance per normalisation parameter. The Sensitivity objective is therefore the direct 2D two-nuisance generalization of the HEP test statistic.

5.5 Nuisance Parameters: Constraints and Non-Analyticity

The penalty terms in Eq. (47) are Gaussian constraints identical in form to the HEP penalty $[(\hat{\beta} - 1)/\sigma_{\text{rel}}]^2$, but with separate widths: $\sigma_{\text{pred}} = 4\%$ (signal flux uncertainty) and $\sigma_{\text{bkg}} = 2\%$ (background uncertainty). These are configured via `analysis/config.json` under `ANALYSIS_UNCERTAINTIES.SENSITIVITY`.

Why no analytic solution exists. In HEP, $e_i = \beta b_i$ under H_0: the single global β factors out and the stationarity condition collapses to the scalar quadratic Eq. (35).

In Sensitivity, $e_{ij} = (1 + A_{\text{bkg}})T_{ij}^{\text{bkg}} + (1 + A_{\text{pred}})p_{ij}$ mixes T_{ij}^{bkg} and p_{ij} in every bin. The two stationarity conditions are:

$$\sum_{ij} p_{ij}\left(1 - \frac{o_{ij}}{e_{ij}}\right) = -\frac{A_{\text{pred}}}{\sigma_{\text{pred}}^2}, \tag{50}$$

$$\sum_{ij} T_{ij}^{\text{bkg}}\left(1 - \frac{o_{ij}}{e_{ij}}\right) = -\frac{A_{\text{bkg}}}{\sigma_{\text{bkg}}^2}, \tag{51}$$

which are jointly non-linear in $(A_{\text{pred}}, A_{\text{bkg}})$ because e_{ij} in the denominators depends on both unknowns. Full 2D minimization is required.

5.6 Barlow-Beeston Mask for 2D Templates

Bins where the background template is zero are excluded:

$$\mathcal{M}_{ij} = \mathbf{1}\left[T_{ij}^{\text{bkg}} > 0\right], \tag{52}$$

passed as `bb_mask` to `Sensitivity_Fitter`. This prevents spurious large deviance from signal-only bins lacking a reliable background model, following the same spirit as HEP's Barlow-Beeston mask (Section 4.6).

5.7 Minimization: scipy L-BFGS-B

The objective Eq. (47) is minimized over $(A_{\text{pred}}, A_{\text{bkg}})$ using the L-BFGS-B algorithm [5, 6]:

$$(\hat{A}_{\text{pred}}, \hat{A}_{\text{bkg}}) = \arg \min_{A_{\text{pred}}, A_{\text{bkg}}} \chi^2(A_{\text{pred}}, A_{\text{bkg}}), \tag{53}$$

with box constraints:

$$|A_{\text{pred}}| \leq 10\,\sigma_{\text{pred}}, \qquad |A_{\text{bkg}}| \leq 10\,\sigma_{\text{bkg}}. \tag{54}$$

L-BFGS-B uses gradient information and is well-suited to smooth, convex objectives with box constraints. Implemented in `lib/lib_root.py:Sensitivity_Fitter.Fit`.

5.8 Oscillation Grid Scan and Cut Quality Score

For each analysis cut $(N_{\text{hits}}, N_{\text{ophits}}, N_{\text{adjcl}})$ and each grid point $\vec{\theta}_k$:

1. Construct the Asimov dataset $o_{ij}(\vec{\theta}_k)$.
2. Minimize χ^2 against p^{solar}: obtain $\chi^2_{\odot}(\vec{\theta}_k)$.
3. Minimize χ^2 against p^{react}: obtain $\chi^2_{\text{react}}(\vec{\theta}_k)$.

The **cut quality score** is the average wrong-hypothesis χ^2 (larger $\Rightarrow$ better discrimination):

$$\text{Score(cut)} = \tfrac{1}{2}\left[\chi^2_{\odot}(\vec{\theta}_{\text{react}}) + \chi^2_{\text{react}}(\vec{\theta}_{\odot})\right]. \tag{55}$$

The resulting χ^2 surface over the oscillation grid gives the sensitivity contours.

5.9 Template Construction Pipeline

The 2D templates T^{sig} and T^{bkg} are assembled from three physical inputs via a two-stage convolution. Figures 13–14 show the nadir time-fraction $p(\cos\eta)$ (earth geometry, fixed by DUNE latitude 44.35°) and the solar survival probability $P(\nu_e \to \nu_e;\, E_\nu^{\text{true}}, \cos\eta)$ computed at the best-fit oscillation parameters. Figures 15–16 present the 1D background spectrum $b(E_{\text{reco}})$ from the selected cuts and the 1D marginal signal spectrum $s(E_{\text{reco}})$ obtained by summing the oscillated signal over nadir bins. Figures 17–18 show the fully assembled 2D templates: T^{bkg} is formed by weighting $b(E_{\text{reco}})$ by $p(\cos\eta)$ via `_project_1d_to_2d`; T^{sig} is the result of the full convolution $T^{\text{sig}} = P_{\nu_e\to\nu_e} \cdot h^T$, where h is the $(E_{\text{reco}} \times E_{\text{true}})$ smearing matrix.

Figure 13: Nadir time-fraction $p(\cos\eta)$ at DUNE latitude (44.35°N), derived from solar ephemeris. This distribution weights each nadir strip in the 2D background template.

Figure 14: Solar-neutrino oscillogram $P(\nu_e \to \nu_e;\, E_\nu, \cos\eta)$ at best-fit parameters ($\Delta m^2_{21} = 6 \times 10^{-5}\,\text{eV}^2$, $\sin^2\theta_{12} = 0.303$). Each row is the survival-probability spectrum for one nadir bin.

Figure 15: 1D background spectrum $b(E_{\text{reco}})$ after optimal cut selection, showing individual components and the smoothed total used as input to the 2D background template.

Figure 16: 1D marginal signal spectrum $s(E_{\text{reco}}) = \sum_\eta T^{\text{sig}}$ at solar best-fit parameters and 30 kt·yr exposure.

Figure 17: 2D background template $T^{\text{bkg}}(\cos\eta, E_{\text{reco}})$: the 1D spectrum $b(E)$ tiled over nadir and reweighted by $p(\cos\eta)$.

Figure 18: 2D signal template $T^{\text{sig}}(\cos\eta, E_{\text{reco}})$: result of convolving the smearing matrix h with the oscillogram $P(\nu_e \to \nu_e)$. The nadir-dependent wiggles encode the MSW day-night modulation exploited by the log-likelihood ratio.

5.10 Results

Figures 19–24 show the Sensitivity analysis outputs for both HD configurations (SolarEnergy, nominal, 30 kt·yr). The top row presents $\sin^2\theta_{12}$ contours for the solar and reactor hypotheses; the bottom row shows significance vs. exposure.

6 Comparison Across Analyses

Table 1 summarizes the statistical model for each analysis.

Narrative progression. The three analyses exhibit a clear progression in statistical complexity centred on the treatment of nuisance parameters.

The Day-Night analysis makes no nuisance-parameter approximations: signal and background are evaluated exactly in two separate exposure intervals, and the asymmetry uncertainty is propagated analytically as a scale factor on the signal. The three-source background uncertainty in the Error Gaussian (Eq. (16)) is the most detailed error propagation of the three analyses, but no LLR profiling is performed.

The HEP analysis introduces the full profile-likelihood framework. The critical observation is that a single global β correlated across all bins leads to a scalar stationarity equation, whose positive root is a closed-form quadratic. This makes the PL computation exact and fast. PAVA post-processing and adaptive rebinning address practical numerical issues (oscillations at low S/B; low-statistics tail bins).

The Sensitivity analysis takes the same Baker-Cousins Poisson deviance as HEP's per-bin LLR, but extends it to 2D and introduces a second nuisance A_{pred} that couples the model both to signal and background simultaneously. This coupling breaks the factorization that enabled the HEP quadratic, requiring a full numerical 2D minimization. The result is a χ^2 map rather than a significance curve.

7 Summary of Significance Outputs

Table 2 lists all significance output columns.

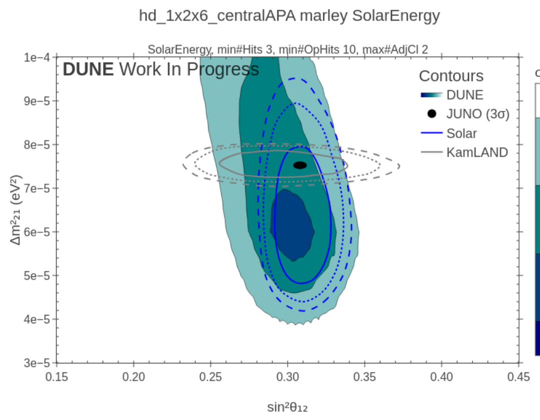

Figure 19: centralAPA: $\Delta\chi^2$ contours in $(\sin^2\theta_{12}, \Delta m^2_{21})$ under the solar oscillation hypothesis.

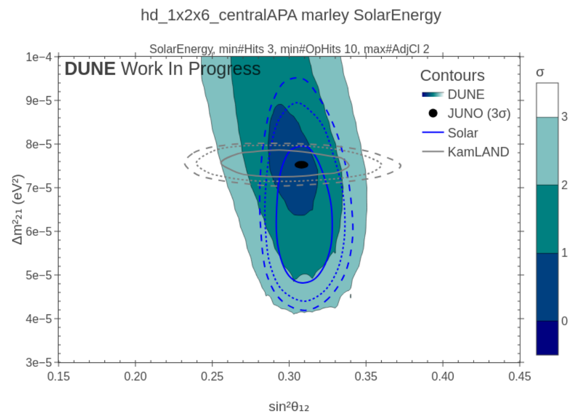

Figure 20: centralAPA: $\Delta\chi^2$ contours under the reactor oscillation hypothesis.

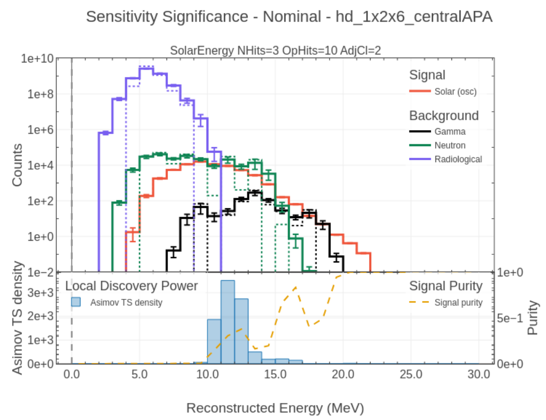

Figure 21: centralAPA: hypothesis-separation significance vs. exposure at 30 kt·yr.

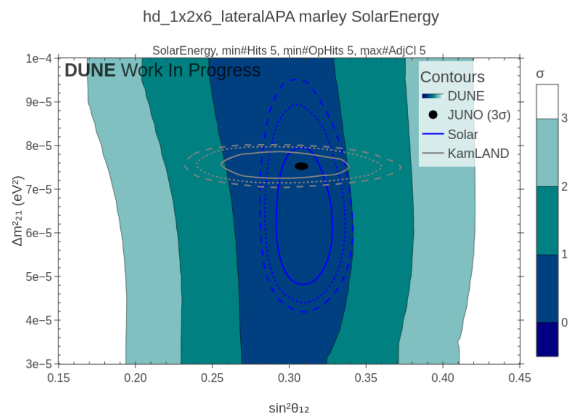

Figure 22: lateralAPA: $\Delta\chi^2$ contours under the solar hypothesis.

8 Workflow Flags and Configuration

Per-analysis workflow feature flags are controlled by `analysis/config.json` under `WORKFLOW.{ANALYSI`

References

References

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[2] S. Baker, R.D. Cousins, *Clarification of the use of chi-square and likelihood functions in fits to histograms*, Nucl. Instrum. Meth. **221** (1984) 437–442. https://doi.org/10.1016/0167-5087(84)90016-4

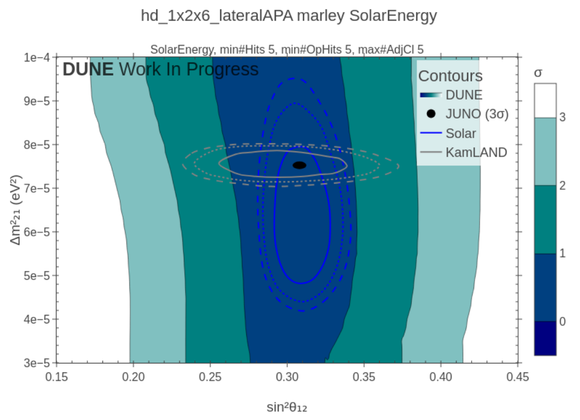

Figure 23: lateralAPA: $\Delta\chi^2$ contours under the reactor hypothesis.

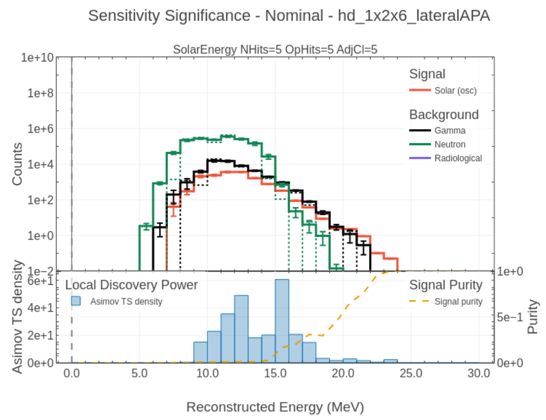

Figure 24: lateralAPA: hypothesis-separation significance vs. exposure.

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[4] T. Robertson, F.T. Wright, R.L. Dykstra, *Order Restricted Statistical Inference*, Wiley, 1988.

[5] R.H. Byrd, P. Lu, J. Nocedal, C. Zhu, *A limited memory algorithm for bound constrained optimization*, SIAM J. Sci. Comput. **16** (1995) 1190–1208.

[6] C. Zhu, R.H. Byrd, P. Lu, J. Nocedal, *Algorithm 778: L-BFGS-B: Fortran subroutines for large-scale bound-constrained optimization*, ACM Trans. Math. Softw. **23** (1997) 550–560.

Table 1: Statistical model comparison: Day-Night, HEP, and Sensitivity.

Feature	Day-Night	HEP	Sensitivity
Observable	1D rate-difference spectrum	1D energy spectrum	2D (energy $\times$ azimuth)
Signal	$s_i{=}\mathcal{E}\xi(r_i^{\text{night}}{-}r_i^{\text{day}})$	$s_i{=}\mathcal{E}r_i^{\text{hep}}$	$T_{ij}^{\text{sig}}(\vec{\theta})$
Goal	Detect day-night asymmetry	Discover hep flux	Map $(\Delta m^2, \sin^2\theta)$ contour
Significance type	Gaussian, two-sample Asimov	Gaussian, Asimov, PL	χ^2 surface
Nuisances	None (error propagated)	$1 \times \beta$ (background)	$A_{\text{pred}} + A_{\text{bkg}}$
Nuisance solution	—	Closed-form quadratic (Eq. 36)	L-BFGS-B (2D)
Penalty form	—	$[(\hat{\beta}{-}1)/\sigma_{\text{rel}}]^2$	$(A_{\text{pred}}/\sigma_{\text{pred}})^2{+}(A_{\text{bkg}}/\sigma_{\text{bkg}})^2$
Per-bin deviance	Eq. (19)	Eq. (37) per bin	Eq. (49)
Effective background	B_i^{eff} (Eq. 12)	$b_i = \mathcal{E}\sum_{c\neq\text{hep}} r_i^c$	T_{ij}^{bkg} (2D template)
Barlow-Beeston mask	No	$N_i^{\text{MC}} \geq N_{\text{MC}}^{\text{min}}$	$T_{ij}^{\text{bkg}} > 0$
Asymmetry/signal band	$\xi \in \{1 \pm \varepsilon_{\text{tot}}, 1\}$	$d \in \{2.9, 3.0, 3.1\}$	—
PL post-processing	—	Gauss(σ=6) + PAVA (optional)	—
Spike filter	—	$\Delta_{\max}$ on pre-PAVA	—
Output	Z vs. exposure	Z vs. exposure	χ^2 map over grid

Table 2: Significance output columns across analyses.

Analysis	Label	Histogram	Rebinning	Estimator	Monoton
Day-Night					
	`RawGaussian`	Raw	None	Gaussian	No
	`RawErrorGaussian`	Raw	None	Gaussian+σ_i^{eff}	No
	`Gaussian`	Smooth	None	Gaussian	No
	`ErrorGaussian`	Smooth	None	Gaussian+σ_i^{eff}	No
	`RawAsimov`	Raw	None	Two-sample LLR	No
	`Asimov`	Smooth	None	Two-sample LLR	No
HEP					
	`RawGaussianNoRebin`	Raw	None	Gaussian	No
	`RawAsimovNoRebin`	Raw	None	Asimov	No
	`RawGaussian`	Raw	Adaptive	Gaussian	Yes
	`RawAsimov`	Raw	Adaptive	Asimov	Yes
	`GaussianNoRebin`	Smooth	None	Gaussian	No
	`AsimovNoRebin`	Smooth	None	Asimov	No
	`Gaussian`	Smooth	Adaptive	Gaussian	Yes
	`Asimov`	Smooth	Adaptive	Asimov	Yes
	`RawProfileLikelihood`	Raw	None	PL	Yes (post
	`ProfileLikelihood`	Smooth	None	PL	Yes (post
	`PreIsotonicProfileLikelihood`	Smooth	None	PL (pre-PAVA)	No
Sensitivity					
	`chi2_solar/react`	2D smooth	None	Baker-Cousins	—

Table 3: Orchestrator flags in `10SensitivityAnalysis.py`.

Flag	Default	Effect when disabled
`--computation`	True	Skip all computation; run only plot macros
`--significance`	True	Skip `12DayNight.py`, `13HEP.py`, `14Sensitivity*.py`
`--fiducialization`	True	Skip `0XFiducializeSignal.py`, `0YBestFiducial.py`
`--rebin`	True	Skip `11AnalysisSignal.py` adaptive rebinning
`--optimization`	False	Run smoothing sigma optimizer `optimize_smoothing.py`

Table 4: Per-analysis workflow feature flags.

Key	Default	Effect when true
DAYNIGHT.background_error	true	Compute Error Gaussian curves (Eq. 14)
DAYNIGHT.significance_bins	true	Save per-bin significance spectra at the display exposure
HEP.pl_signal_bands	true	Evaluate three signal normalizations for $\pm 1\sigma_s$ PL bands
HEP.pl_isotonic	false	Apply Gaussian+PAVA post-processing to PL curves
HEP.significance_bins	false	Save per-bin significance spectra at the display exposure